

What a world model’s uncertainty output actually reports: an independent reproduction of the Robotic World Model

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Abstract

We independently reproduce the proprioceptive dynamics model of Li, Krause and Hutter (*Robotic World Model*, arXiv:2501.10100) and its uncertainty-aware follow-up (arXiv:2504.16680), building the model from scratch on CPU and verifying it against the released reference at the level of outputs, losses and gradients before training anything. The base paper’s central claim reproduces, and by a wide margin: trained autoregressively, the model reaches normalised error 0.3509 at a 368-step horizon on held-out episodes against 1.5540 for teacher forcing, a factor of $4.4\times$, with a bootstrap interval of $[0.56, 2.05]$ excluding zero and the gap positive on all 10 of 10 episodes. That verdict was fixed by a decision rule committed to git before the runs that tested it existed. The follow-up’s uncertainty output does not survive contact with a calibration measurement. The released checkpoint’s predicted standard deviation is **$7,878\times$ smaller than its own mean absolute error**, giving 0.56% coverage at $\pm 1\sigma$ where a calibrated Gaussian gives 68.3%. We show analytically why: the state loss is squared error on a reparameterised sample with no $\log\sigma$ term, minimised at $\sigma = 0$, and the bound term that should oppose this cancels algebraically. We then run the correction — the authors’ own unused `gaussian_nll` branch — and it fails differently rather than succeeding, reaching $10.9\times$ overconfidence. Measuring all four models we trained or scored puts the failure precisely. The teacher-forced arm has the most input-dependent σ ($15.6\times$ the autoregressive arm’s) and the strongest σ -versus-error ordering (45 of 45 dimensions positive, $P = 5.68e-14$), and is still $315\times$ overconfident. These models can learn *which* predictions will be worse. They cannot learn *how wrong* they will be. A downstream user who needs a ranking may be served; one who needs an interval is not, under any of the four. We also report four defects in the released pipeline, evidence that the released checkpoint cannot have come from the released recipe, and five retractions of our own numbered claims — one of which is the finding that one of our own pre-registrations was not, in fact, pre-registered. Every number in this paper is generated from a file in `results/`; none is typed by hand.

1 Introduction

A world model that reports its own uncertainty is more useful than one that does not, and the uncertainty-aware Robotic World Model reports one. This paper asks what that number means.

We came to the question sideways. Our aim was an ordinary reproduction: rebuild the proprioceptive dynamics model from scratch, check it against the released implementation, and see whether the paper’s central training claim holds. It does. But the same rebuild made a second question cheap to ask, because we had a from-scratch model, the released checkpoint, and a harness that could score both: *is the predicted σ calibrated?* It is not, by three to four orders of magnitude, and the reason is structural rather than incidental.

Three things distinguish this from a re-run of the authors’ code.

We rebuilt rather than imported. The forward pass, the loss and the training step are written from scratch and then checked against the reference: outputs match bitwise, and losses

and gradients match to 0.000e+00 across 7 loss terms and 106 parameter tensors before any training begins (Appendix A). A discrepancy found later is therefore a property of the method, not of our wiring.

Decision rules were committed before the data. The verdicts below were fixed in advance, in git, with timestamps a reader can check (§7, Figure 4). One of them returned "cannot be settled" and we report that too.

We retract our own findings when they fail. Four claims in this work are withdrawn on evidence this project produced, and the retractions are kept in the record rather than deleted. One of them concerns the pre-registration discipline itself.

2 Setup

Data. The released dataset is 10,000 rows of ANYmal D proprioceptive state and policy actions at 50 Hz. It is not one recording: it is ten concatenated 20-second episodes, and its termination column is identically zero, so nothing in the file marks the boundaries. The reference window builder therefore marks all 9,961 windows valid, including 352 that splice one episode's end onto the next one's start. The usable, episode-respecting count is 9,609 — 10,000 rows, less 39 that cannot start a full window, less 352 that cross a boundary. The contamination rate is 3.53%.

Model. A GRU-based ensemble predicting the next proprioceptive state, with a mean head and a bounded log- σ head, plus auxiliary heads for contact and termination. The paper describes two loss terms; the implementation has 7.

Evaluation. Two arenas, held separate throughout: *out-of-sample*, the two episodes withheld from training, and *in-sample*, the eight used for it. We report both, because the released evaluation draws its trajectories from training data and the distinction is invisible in the original.

Effective sample size. Trajectory count is not sample size. Two 400-step trajectories whose spans overlap are not independent evidence, and the out-of-sample arena contains only 4 mutually non-overlapping 400-step trajectories. Every interval in this paper is a bootstrap over independent trajectories, and every table reports that count.

3 The base paper's central claim reproduces

Claim under test. Training the dynamics model on its own autoregressive rollouts beats training it with teacher forcing, at deployment horizons.

Rule, committed in advance (commit efc35b8, and it names conditions rather than outcomes). Three conditions, all required: the out-of-sample gap at $h = 368$ excludes zero under a bootstrap over independent trajectories; the sign is consistent across episodes; and the effect survives at 10,000 iterations rather than only at the paper's 2,500.

Result. Every condition holds. At $h = 368$ out-of-sample after 10,000 iterations, autoregressive training reaches **0.3509** against teacher forcing's **1.5540** — a factor of **4.4** $\times$, gap 1.2033, 95% bootstrap interval [0.56, 2.05], on $n = 4$ independent trajectories. The per-episode gap is positive on **10 of 10** episodes.

What does not hold, and we say so. At $h = 8$ — the horizon the model is trained on — the same comparison out-of-sample gives a gap of 0.008 whose interval includes zero. The advantage is a long-horizon phenomenon. An earlier rule of ours, anchored at $h = 8$, returned "cannot be settled"; anchoring a rule to the horizon the claim is actually about was a correction we had to make in advance of the runs, not after them (§7).

The pattern is consistent across the design. Under the correct cluster bootstrap, the out-of-sample gap excludes zero in **4 of 4** long-horizon cells — both trajectory lengths crossed with both checkpoints — and in **0 of 4** at $h = 8$. These figures are relative-L1; the nRMSE aggregation is reported separately and does not change the direction.

4 The uncertainty output is unusable, and that is a property of the objective

4.1 The measurement

For each model we compute the mean predicted σ , the mean absolute realised error, and the fraction of realised errors falling inside $\pm 1\sigma$. A calibrated Gaussian puts 68.3% inside $\pm 1\sigma$.

model	mean error / mean σ	coverage at $\pm 1\sigma$, $h=1$	coverage at $h=368$
faithful Arm A (sampled MSE)	52.2×	11.67%	1.75%
corrected Arm A (<code>gaussian_nll</code>)	10.9×	42.78%	8.57%
teacher-forced Arm B	315×	12.96%	0.56%
released checkpoint	7,878×	0.56%	0.04%

Every model is overconfident by between one and four orders of magnitude (Figure 1).

4.2 Why: the optimum is $\sigma = 0$

The state loss is squared error on a *sample* drawn from the predicted Gaussian, not a likelihood:

$$\mathcal{L} = \mathbb{E}[(\mu + \sigma\varepsilon - y)^2] = (\mu - y)^2 + \sigma^2$$

which is minimised at $\sigma = 0$ for any μ . There is no $\log\sigma$ term to oppose it. The bound term that appears to oppose it does not, because `max_logstd` is not an independent parameter — it is constructed as `min_logstd + exp(log_delta_logstd)`, so

$$\overline{\log \sigma_{\max}} - \overline{\log \sigma_{\min}} = \overline{\exp(\log \Delta_{\log \sigma})}$$

and `min_logstd` cancels algebraically, taking no gradient from that term. The floor the interval closes onto therefore freezes while the interval closes: a one-way ratchet.

We predicted the collapse from this algebra before training, then observed it. Across all 17 runs the collapse is linear in iteration count and its rate is nearly identical (Figure 3a). Under the corrected objective the sign flips (Figure 3b) — which is the strongest evidence that the mechanism is the objective and not the optimiser, the data or the architecture.

4.3 The correction fails differently rather than succeeding

The reference contains an unused `gaussian_nll` branch. Running it reverses the collapse and improves the magnitude from 52.2× to 10.9× overconfident. It does not produce a usable estimate, and it destroys something the faithful arm had: the σ -versus-error ordering falls from 39/45 dimensions positively correlated ($P = 5.42\text{e-}07$) to 21/45 ($P = 7.66\text{e-}01$, chance).

4.4 The failure is one of magnitude, not of ordering

Measuring the teacher-forced arm — which we had trained for §3, and which our own first three calibration tables omitted — sharpens the finding:

model	σ variation across inputs (CoV)	dims with $r(\sigma, \text{error}) > 0$	P
faithful Arm A	0.0076	39/45	5.42e-07
corrected Arm A	0.0059	21/45	7.66e-01
teacher-forced Arm B	0.1188	45/45	5.68e-14
released checkpoint	0.0177	20/45	5.51e-01

Arm B’s σ is $15.6\times$ more input-dependent than the faithful arm’s, and its ordering is the strongest of the four by a wide margin (mean $r = 0.257$). It is still $315\times$ overconfident.

So σ collapsing to a constant is a property of the *autoregressive* arms and the released checkpoint, not of the objective in general — and input-dependence and correct ranking are both achievable without the interval becoming meaningful. **The failure is specifically magnitude calibration.**

4.5 The structural excuse does not survive

One could argue that a model trained on an 8-step horizon cannot be expected to report calibrated uncertainty about step 368. It cannot report it about step 8 either. Inside the trained horizon, σ is flat while error grows (Figure 2):

model	σ growth, step 1 $\rightarrow$ 8	error growth, step 1 $\rightarrow$ 8
faithful Arm A	$0.9241\times$	$3.49\times$
corrected Arm A	$1.0003\times$	$3.41\times$
teacher-forced Arm B	$1.0096\times$	$6.11\times$
released checkpoint	$1.0007\times$	$1.79\times$

The faithful arm’s σ *declines* ($0.9241\times$) while its error grows $3.49\times$. The coverage collapse in Figure 1(b) is therefore driven entirely by growing error against a fixed σ .

5 Defects in the released pipeline

5.1 Ten unmarked episode boundaries. §2. The window builder reads a termination column that is identically zero, so it marks all 9,961 windows valid.

5.2 Training and evaluation disagree on action alignment, and evaluation is the broken one. Row t holds the action that *produced* state t . The training path pairs states and actions index-for-index, which is causally correct. The evaluation path feeds the action from $t-1$ to predict state t — stale by one step. Scored correctly the released checkpoint is materially better than its own released evaluation reports.

5.3 No held-out evaluation. Evaluation trajectories are drawn from training data. For the released checkpoint, trained on the entire file, no held-out measurement is possible at all.

5.4 What the spliced windows cost: nothing measurable. We trained a contaminated arm on 7,882 windows — the clean 7,687 plus 195 splices — and, because that confounds

content with *count*, a duplication control adding the same 195 windows as exact copies of windows already present.

The arm’s contamination rate is 2.47%, against the reference pipeline’s 3.53%. It is deliberately lower: we splice only boundaries whose *both* sides are training episodes, because four of the reference’s nine put held-out rows into training. That is a leakage problem rather than a physics one, and including it would have invalidated our own comparison. So this experiment measures the cost of training on physically impossible transitions, and not the reference’s full exposure.

Training loss over the final 250 iterations: duplication costs 0.90%, splicing costs 21.57%. The bootstrap interval on duplicated – clean is $[-0.0061, 0.0215]$, including zero. So the rise is caused by splice content, not by dataset size — a control we ran only because the first version of this finding inferred the mechanism without it.

In rollout, across 32 cells (two arenas $\times$ two trajectory lengths $\times$ two checkpoints $\times$ two horizons $\times$ two metrics), contamination hurts in **0** of 32 and helps in 9 (Figure 5a). The control is inert, differing from clean in 2 cells. **The unmarked boundaries remain a real defect on leakage grounds; what is now measured is that the physically-impossible-transition component costs nothing detectable at this rate.**

6 The released checkpoint cannot have come from the released recipe

The collapse rate is a clock. Fitting it across our runs and extrapolating to the released checkpoint’s σ state implies **153,270** optimisation steps at the configured learning rate. The refit from our 10,000-iteration runs gives 158,319 and 158,003, spreading 3.3% across the three fits — a linear extrapolation validated over a fourfold extension.

The released configuration says 500 iterations. The paper says 2,500. The checkpoint is tagged 5,000. A second, independent parameter on a slower gradient path implies the same order. And under `gaussian_nll` the implied count is *negative*, which identifies the branch the checkpoint was trained with.

7 Method

An append-only ledger. Every claim in this work has a permanent identifier, an evidence class (source, data, run, external, inference) and a status, in `FINDINGS_LEDGER.md` (154 entries). Claims are never edited in place. A claim that turns out to be wrong is marked superseded, with a pointer to what replaced it, and kept.

Pre-registration, and one failure of it. Decision rules were committed to git before the data that tested them — with one exception, which we report below. Figure 4 shows the lead time for each, computed from commit timestamps: the A/B rule by 1.3 hours, the flip-pattern rule by 4.8 hours, the difficulty-bias rule by 5 minutes, the long-horizon rule by 2 minutes.

The fifth bar is negative. The rule for the duplication control (§5.4) was stated in conversation before the runs but reached git **2.9 hours after the runs finished**, and we found this only by auditing our own git log. The measurement stands — the arm was built and run without reference to its outcome — but the claim that it was pre-registered does not, and we withdraw it. We report it because a discipline that is only checked when it succeeds is not a discipline.

Five retractions on our own evidence, out of 13 superseded claims kept in the record. In

order: a premise about forecast decay that turned out not to exist in the code; a framing of the released checkpoint as "clearly informative" that rested on an $n=10$ estimate we ourselves showed to be biased low; an aggregation artifact that inverted a published-model comparison in our favour, withdrawn when the gating checks we had written refuted it; a per-dimension comparison that turned out to be unmatched; and the claim that σ is input-independent "in all four models", made against a table holding three. The pre-registration claim above is a sixth, retracting a framing rather than a number.

A statistic that was resampling the wrong unit. Our bootstrap pooled three training seeds over a shared set of evaluation trajectories and resampled the pooled vector, while reporting the independent-trajectory count. Each trajectory appeared three times. Resampling trajectories correctly — carrying all seeds with each draw — widens intervals by a mean factor of $1.42\times$ (range 0.96–1.69) and changes 1 of 16 verdicts, in an $h = 8$ cell already recorded as unresolvable. Every long-horizon verdict survives. Both units are reported.

Reproducibility. `./reproduce.sh --quick --force` regenerates 19 artifact files and 4,804 numeric values from a clean clone, 4,804 of them bitwise identical (100.00%), 0 differing. Timing fields and one whole host-measurement artifact are excluded and reported separately.

8 Limitations

Effective sample size bounds every long-horizon claim. The out-of-sample arena has 4 independent 400-step trajectories. That is the binding constraint on §3, and no amount of trajectory oversampling changes it.

Ensemble size. Our main experiment runs at ensemble size 1 against the reference's 5, for CPU budget. The epistemic component of the released model's uncertainty is therefore not reproduced; §4 concerns the aleatoric head, which is per-member.

One dataset, one gait, one terrain. All commands are drawn from one bounded box and the gait is a single trot throughout. "Generalisation" here means across velocity commands, not across gaits or terrain.

Two of our headline analyses rest on a single training seed, because only one 10,000-iteration run per arm exists. This is recorded in the artifacts themselves.

We did not reproduce the policy-learning results of either paper. This is a dynamics-model reproduction only.

9 Conclusion

The Robotic World Model's central training claim reproduces, and the margin is large. Its uncertainty output, in the follow-up that adds one, does not report what a reader would take it to report: the released checkpoint's σ is $7,878\times$ smaller than its own error, and the cause is that the objective's optimum is $\sigma = 0$ with the term that should prevent this cancelling out of the gradient.

The more useful finding came from measuring a model we had already trained for another purpose and had not put on a calibration table. Ranking and input-dependence are achievable — the teacher-forced arm has both — while the interval remains meaningless. Uncertainty in this family of models should be reported as an ordering, or fixed at the objective, but not read as a scale.

10 Data and code

<https://github.com/joyjeet-singh/rwm> — every number above cites a file under `results/`, and `FINDINGS.LEDGER.md` carries the full claim record including the retractions. Neither upstream repository is redistributed; `setup.sh` fetches both at pinned commits and verifies two SHA-256 hashes.

11 References

1. C. Li, A. Krause, M. Hutter. *Robotic World Model: A Neural Network Simulator for Robust Policy Optimization in Robotics*. arXiv:2501.10100, 2025.
2. C. Li, A. Krause, M. Hutter. *Uncertainty-Aware Robotic World Model Makes Offline Model-Based Reinforcement Learning Work on Real Robots*. arXiv:2504.16680, 2025.

A verification chain

What every downstream number rests on. Each level was passed before the next was attempted.

level	claim	result
shapes	parameter counts match the reference	exact
wiring	inference outputs match the reference module	0.000e+00 , bitwise
indexing	the harness feeds the actions it claims	bitwise against the raw CSV
residual	the zero-delta model is the hold-last floor	1.192e-07
objective	losses and gradients match	0.000e+00 across 7 terms, 106 tensors
trainer	can memorise a single batch	1,506× loss reduction

B reproducing

```
./setup.sh # clone upstreams at pinned commits
python3.11 -m venv .venv && . .venv/bin/activate
pip install -r requirements.txt
./reproduce.sh --quick --force # everything except training
```

`--force` matters: a clean clone already contains each stage’s declared output, so without it every stage skips. Training stages are excluded by `--quick`; a full run is roughly 22 hours on two CPU cores.

C Figures



